

Keyush Nisar

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Education

B.Tech in Artificial Intelligence and Machine Learning

Aug 2023 – 2027

D.J. Sanghvi College of Engineering, Mumbai University

Experience

Machine Learning Intern — CerebralZip (VJTI-incubated Startup, Mumbai)

Jul 2026 – Present

- Developing and deploying ML systems for a government environmental and safety monitoring platform, including multilingual incident classification, multi-issue detection, automated department routing, and real-time alert generation from noisy citizen reports.
- Developed time-series forecasting and source-attribution models for environmental monitoring, with evaluation and inference pipelines supporting government-facing deployments and operational alert workflows.
- Prototyped ALPR and computer-vision pipelines for degraded field footage, combining license-plate detection and OCR under low-light, blur, and long-distance conditions to support enforcement workflows.

MLE Research Intern — RAI Tracker (Responsible Systems Ltd) (Remote, London, UK)

Apr 2026 – Jul 2026

- Conducted research on behavioral drift in multi-agent RAG systems, treating runtime telemetry as primary evidence for benchmarking deployed AI behavior across company-defined ethical dimensions.
- Designed a three-gate validation framework – instrumentation validity, statistical validity, and evidentiary independence – that a candidate telemetry signal must pass before being certified as evidence of an ethical-risk pattern.
- Built hierarchical OpenTelemetry instrumentation, a system-agnostic telemetry/drift schema, and interactive dashboards that turned runtime traces into experimental verdicts; continuing the work as a collaborative research manuscript.

AI Production Engineering Intern — EnsanAI (Remote, Delaware, US)

Aug 2025 – Apr 2026

- Designed a domain-specific healthcare AI evaluation framework covering HIPAA-oriented compliance, empathy, behavioral/data drift, and safety metrics to systematically assess model behavior across clinical workflows.
- Deployed production AI microservices on Google Cloud Platform, stress-tested inference workloads, and built an MCP server connecting database/operations tooling with downstream microservices; reduced inference costs by 35% and system latency by 40%.

Projects

Uncertainty-Preserving Distillation for Diffusion World Models |

GitHub

PyTorch, Diffusion Models, MBPO, SAC

- Proved that single-latent teacher–student value-disagreement matching does not uniquely identify **epistemic uncertainty**; constructed counterexamples and derived a multi-latent objective that separately constrains epistemic and aleatoric components.
- Validated the claim across controlled and live SAC experiments; after tracing and fixing a training-pipeline bug, confirmed that the corrected objective recovers both uncertainty channels and partially transfers to policy-driven settings.
- Currently testing whether improved uncertainty fidelity translates into measurable policy-return gains on a harder pre-registered benchmark.

LM-JEPA for Symbolic Regression | PyTorch, Deep Learning, Representation Learning

GitHub

- Researched and adapted Yann LeCun’s Joint Embedding Predictive Architectures (JEPA) from vision domains into symbolic regression.
- Developed a representation-learning pipeline to extract mathematical structure from complex data without relying on autoregressive generation.

Voice-First Financial Agent | FastAPI, LangGraph, OpenTelemetry

GitHub

- Engineered a low-latency WebSocket-based voice agent with asynchronous streaming and VAD-driven interruption handling, achieving sub-500ms time-to-first-audio.
- Built OpenTelemetry-based observability for end-to-end latency and state-transition analysis across multi-stage agent workflows.
- Implemented asynchronous tool and state orchestration to support responsive financial-agent interactions while preserving traceability across agent transitions.

Research & Publications

VRAG: Hierarchical Semantic Retrieval for Long-Form Video Question

2026

Answering | Springer – The Visual Computer (Under Peer Review)

- Contributed to a 3-level VideoRAG framework (**Topic** → **Scene** → **Keyframe**) for coarse-to-fine evidence localization in long-form videos.
- Worked on multimodal and temporal embeddings and the Gaussian-smoothed keyframe scoring method, including its mathematical formulation for temporally stable retrieval.
- Evaluated across 463 videos and 1,103 queries on YouCook2, achieving **42.2% Video R@1** and **57.0% Scene R@3**, with a **3.7× recall gain** over flat retrieval.

DIO: Black-Box Online Routing for Heterogeneous vLLM Replicas |

2026

Springer – The Journal of Supercomputing (Under Review)

- Led the design of a lightweight OpenAI-compatible gateway for stock vLLM using dual-timescale NLMS-style online routing to learn heterogeneous replica performance from black-box end-to-end telemetry.
- Developed an O(1) online latency model that learns per-replica service-time slopes and adaptively routes requests across heterogeneous inference workers.
- Achieved **48.3% lower p99 latency** than Round-Robin under controlled $2\times$ service-time asymmetry, with only $\sim 14\mu s$ control-plane scheduling overhead.
- Identified a key limitation of learned routing: strong relative ranking can coexist with poorly calibrated absolute latency thresholds, motivating a separation between routing and admission decisions.

Skills

GenAI, Governance & Research: RAG, Transformers, PEFT, LoRA, Uncertainty Quantification, Trustworthy ML, LLM Evaluation, LangChain, LangGraph, Guardrails AI, Giskard

Languages & Frameworks: Python, Go, SQL, PyTorch, TensorFlow, Scikit-learn, FastAPI

Systems & Pipelines: vLLM, OpenTelemetry, Kafka, gRPC, Docker, Kubernetes, CI/CD, PostgreSQL

Cloud & MLOps: AWS, GCP, Arize Phoenix, Model Monitoring, Prometheus, Grafana

Co-curricular Activities

Technical Head, ML Mentor & Co-founder — CodeAI (ML Club at DJSCE),
Mumbai

Jun 2025 – Present

- Mentor students in Machine Learning and Data Science and lead technical projects within the club.
- Directed a team developing an AI engine for Aspira Pathology Lab for automated prescription interpretation, achieving 92%+ accuracy across a 3,500+ test catalog.

Awards & Achievements

- **Nexus Kaggle Competition:** 2nd Runner-up – F1 score 99.56% in drug status prediction.
- **Hack2Infinity:** Best in AI – Mental Health App “MindCompanion”.
- **CodeShastra 12th Edition:** 2nd Runner-up – ERP-based AI FaithTech App.
- **Catnip Infotech Hackathon:** Runner-up – Recommendation System.